

Effective Visual Feedback for Virtual Grasping

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ABSTRACT

The purpose of this study is to examine effectiveness of visual feedback in virtual grasping with a bare hand. We propose two types of visual feedback that can generate a pseudo-haptics and compare them with five types of existing visual feedback. In an experiment, a subject performs a pick-and-drop task, in which a virtual object, which is a sphere with a radius of 5 cm, is grasped, moved and released toward a target. Quantitative and qualitative evaluation by twenty subjects revealed that the one type of the proposed visual feedback helped the subjects perform the task most accurately and the other type of the proposed visual feedback was preferred most in the other visual feedback.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in interaction design**; **Virtual reality**.

KEYWORDS

Virtual grasping, Visual feedback, Pseudo-haptics

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1 INTRODUCTION

One of the interactions with virtual objects is virtual grasping, in which the user grasps a virtual object. In virtual grasping, if the fingers are bent deeply to grasp a virtual object, the accuracy and speed when releasing the virtual object toward a target is degraded and fatigue increases. These problems can be reduced by grasping the virtual object with light touch, and visual feedback can change the way the fingers close. There are two basic hand representation ways for virtual grasping: Inner Hand (IH) and Outer Hand (OH) (explained in Section 2.1). In general, IH is easy to find the positions of the actual fingers, but it is not preferred because of its non-realistic appearance. On the other hand, OH is preferred because of its realistic appearance, although the actual fingers tend to penetrate the virtual object deeply because their movements are not visible. To mitigate this trade-off between IH and OH, Prachyabrued et

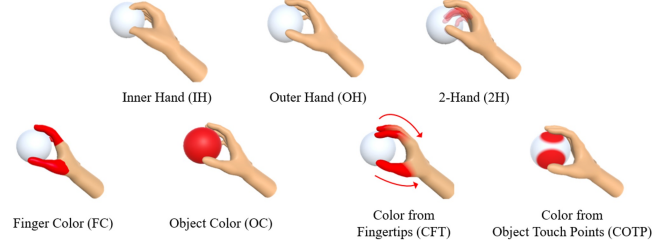


Figure 1: Visual feedback. We proposed CFT and COTP.

al. [Prachyabrued and Borst 2016] presented some visual feedback including Finger Color (FC) and Object Color (OC) (explained in Section 2.1) and evaluated them in terms of task performance and subjective preference. They found that FC and OC were the most preferred, but the task performance still needed examining in detail.

We propose visual feedback by which a pseudo-haptics is generated so that it is easy for the user to understand how much force the user applies for each finger. In the experiment, we examine effectiveness of visual feedback in virtual grasping with a bare hand. Five types of existing visual feedback and two types of proposed visual feedback are evaluated quantitatively and qualitatively.

2 EXPERIMENT

The task was to grasp a virtual object, which was a sphere with a radius of 5 cm, and to move and release it toward a target. We presented seven types of visual feedback while the subject grasped the virtual object with a bare hand, as shown in Figure 1. To track the hand motion, a Leap Motion Controller2 (Ultraleap) was attached to the front of A VIVE Pro (HTC).

To measure the task performance of virtual grasping, the subject performed a pick-and-drop task. The subject was asked to complete the task as quickly and accurately as possible. For each visual feedback, the subject performed the pick-and-drop task eight times. We conducted quantitative and qualitative evaluation on the visual feedback under performing the task.

2.1 Stimuli

We proposed two types of visual feedback that can generate a pseudo-haptics and compared them with five types of existing visual feedback [Canales et al. 2019; Prachyabrued and Borst 2016].

Existing visual feedback:

Inner Hand (IH): The virtual fingers always move in the same way as the actual fingers, so that the virtual fingers are easily embedded in the virtual object.

Outer Hand (OH): The virtual hand stays outside the virtual object while the positions of the actual fingers are inside the virtual object.

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2-Hand (2H): 2H is the combination of IH and OH. In addition to OH, the virtual fingers that follow the actual finger movements are presented in translucent red.

Finger Color (FC): In addition to OH, according to the depth at which each actual finger enters the virtual object, the color of the corresponding virtual finger continuously changes from the skin color to red.

Object Color (OC): In addition to OH, according to the depth at which the actual finger enters deepest the virtual object, the color of the virtual object continuously changes from white to red.

Proposed visual feedback:

Color from Fingertips (CFT): CFT differs from FC in how the color change spreads. According to the depth at which each actual finger enters the virtual object, the color change gradually spreads from the tip to the base of the corresponding finger. The gradual spreading of color from a touch point was used as a means of generating a pseudo-haptics.

Color from Object Touch Points (COTP): COTP differs from OC in how the color change spreads. According to the depth at which each actual finger enters the virtual object, the color change gradually spreads from the touch point of the corresponding finger to the spherical surface with a 1/6 radius. Unlike OC, the depth of each finger can be confirmed, which may improve task performance.

2.2 Evaluation

Task performance below was measured in the pick-and-drop task.

Grasping time: The time taken to grasp the sphere.

Release time: The time taken to release the sphere.

Error: The horizontal distance between the center of the sphere and the center of the target.

Average depth of fingers: The average depth of 5 fingers just before releasing the sphere.

For each visual feedback, the subject answered the ease of grasping (Q1), the ease of releasing (Q2) and the feeling of applying force (Q3) on a 7-point Likert scale from -3 to 3. After all visual feedback were performed, the subject ranked them in terms of Q1, Q2, Q3, “appearance preference (Q4)” and “overall preference (Q5).”

Twenty-two university students (the mean age 22.8, all male) participated in the experiment.

3 RESULTS

To indicate significant differences between the visual feedback, the Friedman test followed by a multiple comparison using the Holm method was performed. The ranks from the 1st to the 7th were converted from 7 points to 1 point, respectively.

Figure 2 shows some of remarkable results. Regarding the grasping time in Figure 2(a), significant differences were found between IH and all other visual feedback except 2H ($p < .05$), and between 2H and all other visual feedback except IH ($p < .05$). The visual feedback in which the virtual fingers always followed the actual fingers (IH, 2H) helped the subjects grasp the virtual object quickly. Because the subjects had no visual constraints when touching the virtual object, they could grasp the virtual object without hesitation.

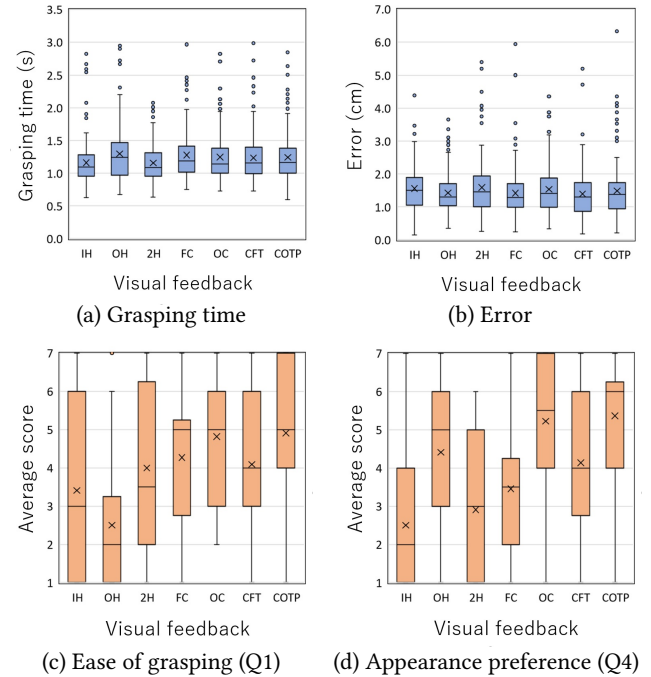


Figure 2: Results of the experiment.

As for the error in Figure 2(b), significant differences were found between IH and CFT ($p < .05$), and between 2H and CFT ($p < .05$). The error for CFT was the least in all visual feedback, when the subjects released the virtual object toward the target. The subjects commented, “Compared to FC, it is easy to see how much force each actual finger is putting on the virtual object,” and “I feel as I am grasping and controlling the virtual object with my fingertips.” The ease of knowing which actual finger was entering the virtual object how much had the greatest impact on accuracy. Regarding the Q1 ranking in Figure 2(c), significant differences were found between OH and OC ($p < .01$), and between OH and COTP ($p < .05$). The visual feedback in which the virtual object changed its color (OC, COTP) were perceived as easier to grasp. For OC and COTP in the Q3 ranking, the subjects strongly perceived the pseudo-haptics, suggesting that the pseudo-haptics facilitated the ease of grasping. As for the Q4 ranking in Figure 2(d), significant differences were found between IH and OH ($p < .01$), IH and OC ($p < .01$), IH and COTP ($p < .05$), 2H and COTP ($p < .05$), and FC and COTP ($p < .05$). Because COTP was preferred most in the Q4 and Q5 rankings and received high ratings overall in the 7-point Likert scale evaluation, changing the color of the virtual object was preferred over changing the color of the virtual fingers.

We conclude that IH or 2H is effective for tasks that require quickness, CFT is effective for tasks in which accuracy is important, and COTP is effective for tasks that focus on user experience.

REFERENCES

- Ryan Canales, Aline Normoyle, Yu Sun, Yuting Ye, Massimiliano Di Luca, and Sophie Jörg. 2019. Virtual Grasping Feedback and Virtual Hand Ownership. In *ACM Symposium on Applied Perception 2019*. Article 4, 9 pages.
- Mores Prachyabrued and Christoph W. Borst. 2016. Design and Evaluation of Visual Interpenetration Cues in Virtual Grasping. *IEEE Transactions on Visualization and Computer Graphics* 22, 6 (2016), 1718–1731.